TRANSFORMATIONS · SUMMARY

How linear maps combine

Every operation here produces another linear map — that closure is what lets the set of all linear maps be a vector space in its own right, rather than merely a collection.

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OPERATIONS

UNCONDITIONAL 2

iv Scalar multiple $(cT)(\mathbf{v}) = c \cdot T(\mathbf{v})$

UNCONDITIONAL

REQUIRES any scalar c

Scaling a linear map keeps it linear, including at $c = 0$, which gives the zero map — the additive identity of $\mathcal{L}(V, W)$. Together with the sum above, this is the second of the two operations a vector space needs.

vi The zero vector is fixed $T(\mathbf{0}) = \mathbf{0}$

UNCONDITIONAL

REQUIRES T linear

COMMON ERROR treating this as a definition — it is a consequence, and its failure is a fast disproof

Follows from $T(0 \cdot \mathbf{v}) = 0 \cdot T(\mathbf{v})$. Useful in the contrapositive: any map sending $\mathbf{0}$ somewhere else is not linear, no further work needed. This is why an affine map $\mathbf{x} \mapsto A\mathbf{x} + \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$ fails immediately.

> **FAST DISPROOF**
>
> ```
> T(x, y) = (x + 1, y)
> T(0, 0) = (1, 0) ≠ (0, 0) → not linear
> ```

CONDITIONAL 4

i Composition $(S \circ T)(\mathbf{u}) = S\big(T(\mathbf{u})\big)$

STAYS LINEAR

HOLDS WHEN the codomain of T matches the domain of S

Apply T's linearity inside S's and the result falls out. In matrix terms this is exactly matrix multiplication — which is why that product is defined the way it is, and why it fails to commute: $S \circ T$ and $T \circ S$ are different maps.

ii Inverse $T^{-1} \circ T = I, \quad T \circ T^{-1} = I$

STAYS LINEAR

HOLDS WHEN T is a bijection

FAILS WHEN T has nontrivial kernel or misses part of the codomain

The inverse of a linear bijection is itself linear — not obvious, but true. An invertible map between spaces of equal dimension is an isomorphism, meaning the two spaces are structurally the same object written twice. See the invertibility equivalence for the matrix form of the condition.

iii Sum $(S + T)(\mathbf{v}) = S(\mathbf{v}) + T(\mathbf{v})$

STAYS LINEAR

HOLDS WHEN S and T share domain and codomain

Defined pointwise, and linear because both summands are. This is what supplies the addition on $\mathcal{L}(V, W)$ — the operation exists because linear maps are closed under it, not by stipulation.

v The space of linear maps $\dim \mathcal{L}(V, W) = \dim(V) \cdot \dim(W)$

CONDITIONAL

HOLDS WHEN both V and W finite-dimensional

Closure under sum and scalar multiple makes $\mathcal{L}(V, W)$ a vector space whose elements happen to be functions. Its dimension is the product of the two, which is the same count as the entries of an $m \times n$ matrix — the matrix representation identifies it with the space of $m \times n$ matrices.

Read downward and the entries build on each other: composition and inversion give $\mathcal{L}(V, V)$ the structure of an algebra, while sum and scalar multiple give $\mathcal{L}(V, W)$ the structure of a vector space. The last entry is the two facts combined.